

EARLY AND ACCURATE RECESSION DETECTION USING CLASSIFIERS ON THE ANTICIPATION-PRECISION FRONTIER

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May 2025

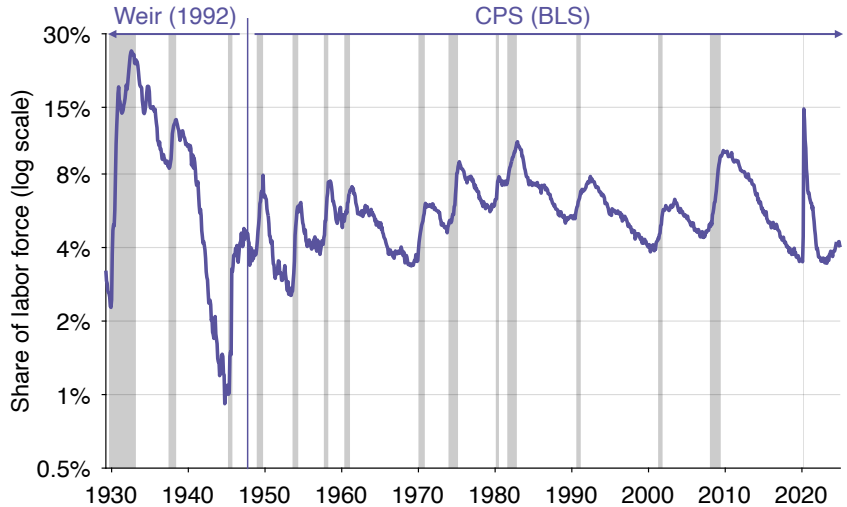
Available at <https://pascalmichailat.org/17/>

DETECTING US RECESSIONS IN REAL TIME

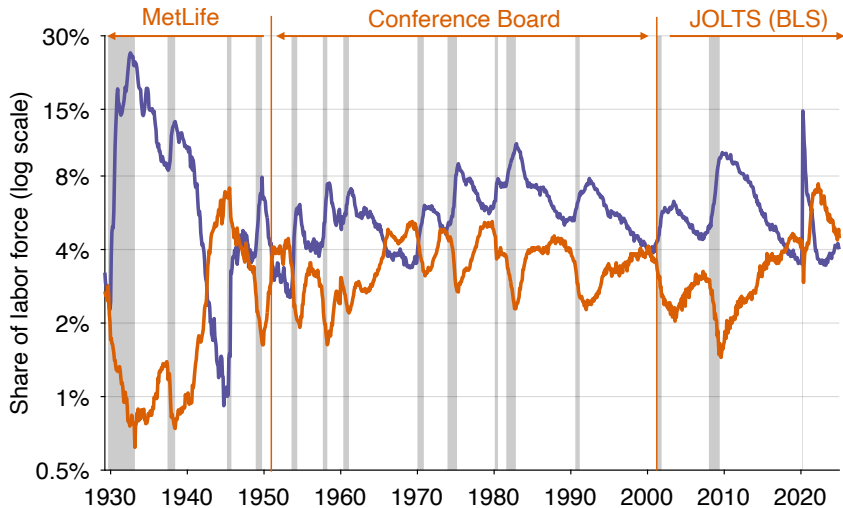
- Threshold rules + labor market data work best
 - Horserace by Crump, Giannone, Lucca (2020)
- Popular detection rule: Sahm (2019)
 - Based on the unemployment rate
 - No false positives, no false negatives on 1960–2021
- Faster and more robust rule: Michailat & Saez (2025)
 - Based on the unemployment & vacancy rates
 - Average detection delay: 1.2 months < 2.7 months
 - No false positives, no false negatives on 1929–2021
- While threshold is chosen optimally, data filtering is arbitrary
- Here: develop optimal detection method from labor market data

US UNEMPLOYMENT & JOB VACANCIES, 1929–2025

US UNEMPLOYMENT RATE (PETROSKY-NADEAU, ZHANG 2021)



US VACANCY RATE (PETROSKY-NADEAU, ZHANG 2021)



CONSTRUCTING RECESSION INDICATORS

1. Smoothing data by moving average ($0 \leq \alpha$)
 - a. Simple: $\bar{u}(t) = \left[\sum_{j=0}^{\alpha} u(t-j) \right] / (\alpha + 1)$
 - b. Exponentially weighted: $\bar{u}(t) = \alpha u(t) + (1 - \alpha) \bar{u}(t-1)$
2. Detecting turning points ($1 \leq \beta$):

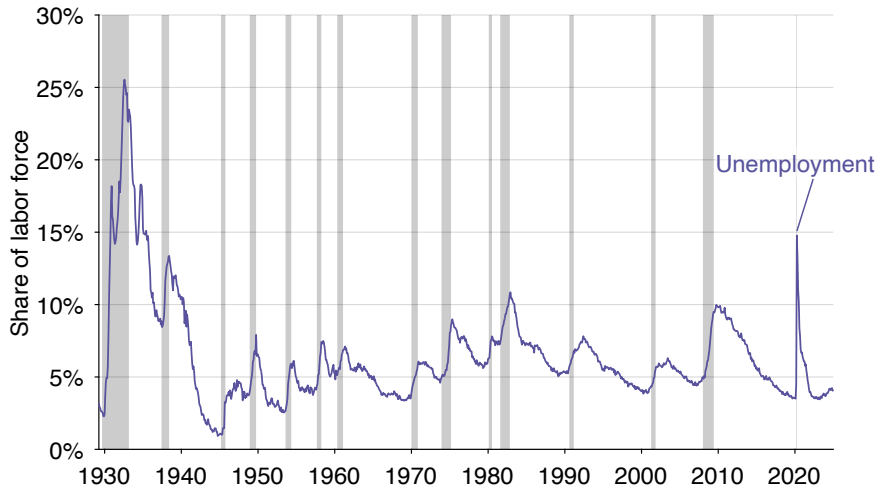
$$\tilde{u}(t) = \bar{u}(t) - \min_{0 \leq j \leq \beta} \bar{u}(t-j)$$

3. Curving data changes ($0 \leq \gamma \leq 1$):

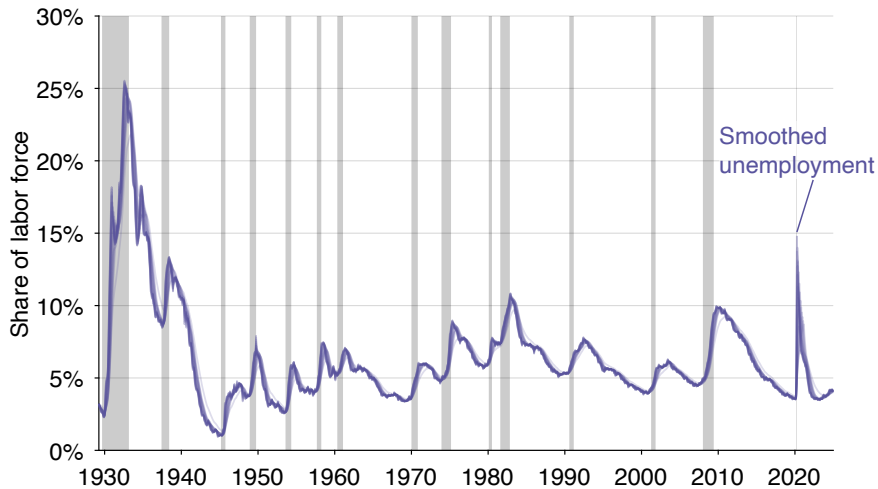
$$\hat{u}(t) = \frac{\tilde{u}(t)}{\bar{u}(t)^{\gamma}}$$

4. Combining indicators ($0 \leq \delta \leq 1$)
 - a. Unemployment-vacancy: $i(t) = \delta \hat{u}(t) + (1 - \delta) \hat{v}(t)$
 - b. Min-max: $i(t) = \delta \min(\hat{u}(t), \hat{v}(t)) + (1 - \delta) \max(\hat{u}(t), \hat{v}(t))$

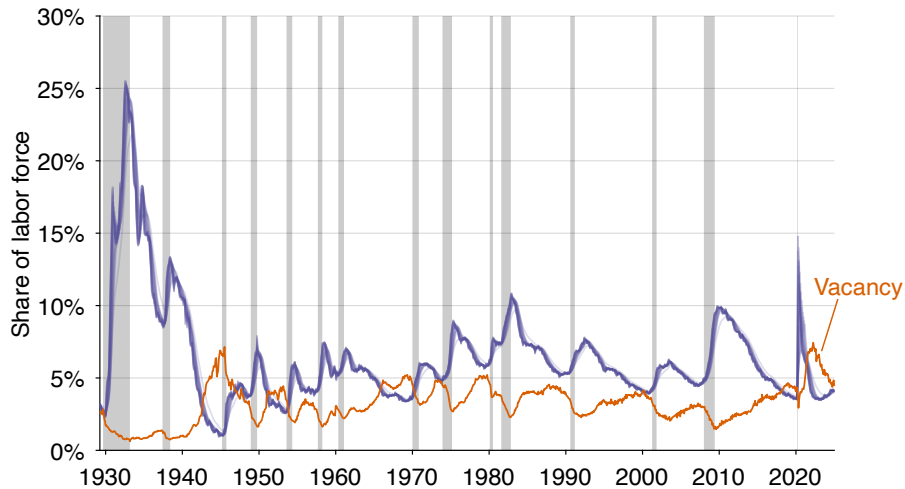
SMOOTHING UNEMPLOYMENT



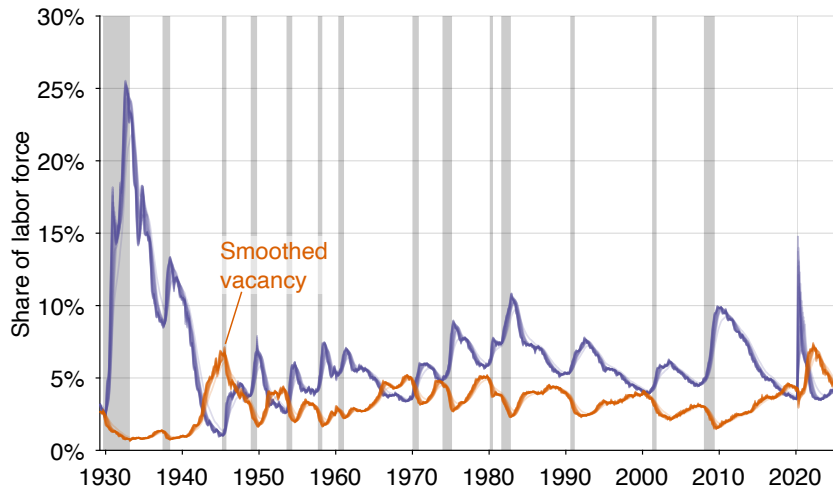
SMOOTHING UNEMPLOYMENT



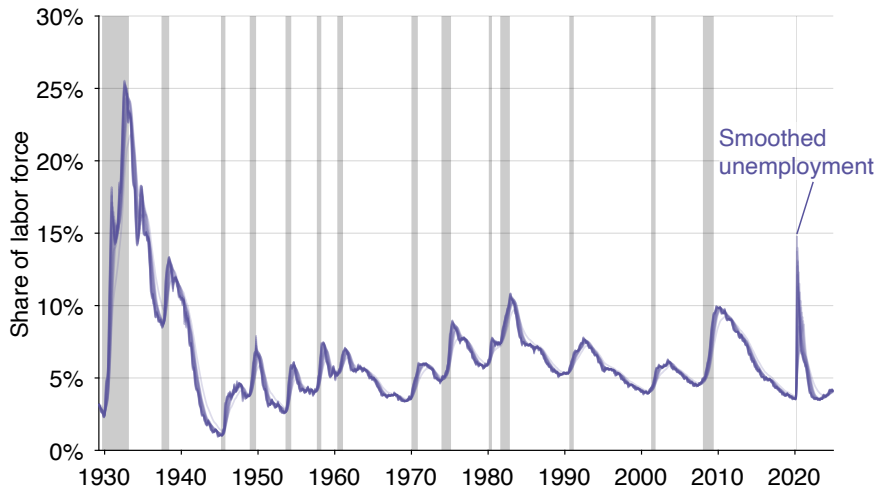
SMOOTHING VACANCY



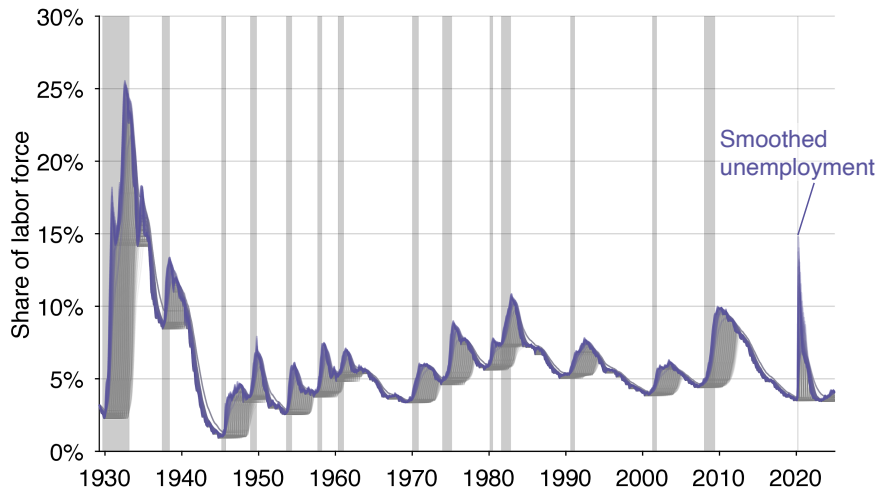
SMOOTHING VACANCY



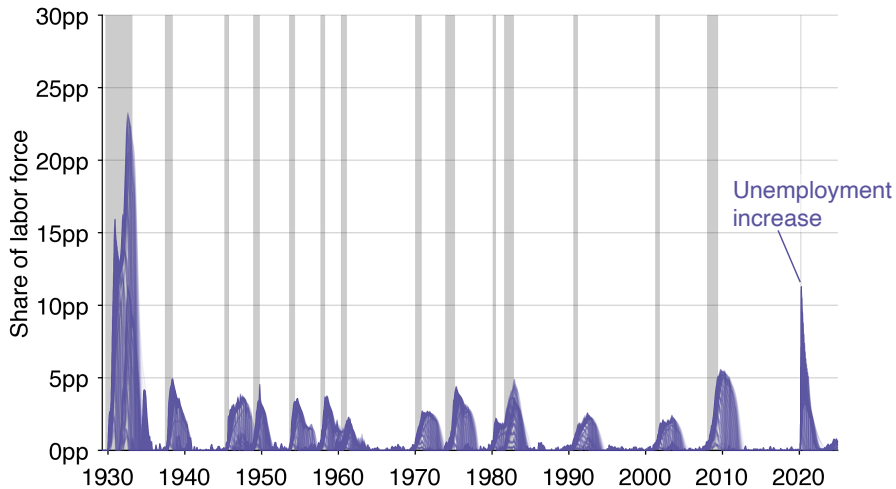
DETECTING UNEMPLOYMENT TURNING POINTS



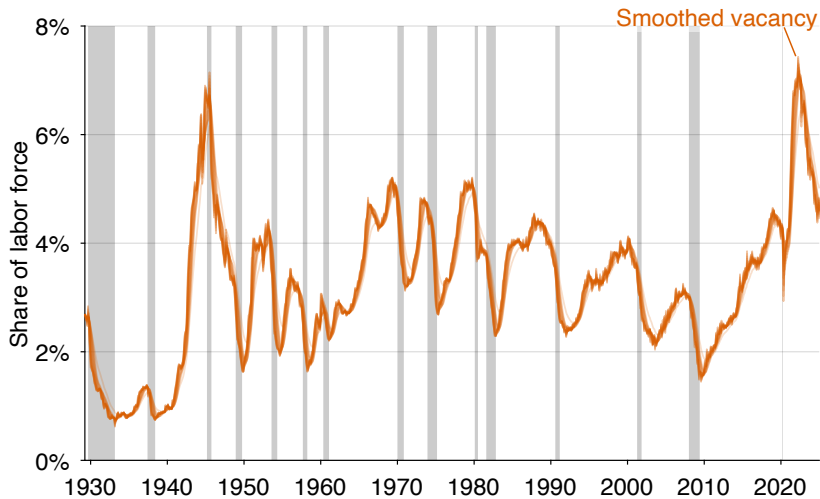
DETECTING UNEMPLOYMENT TURNING POINTS



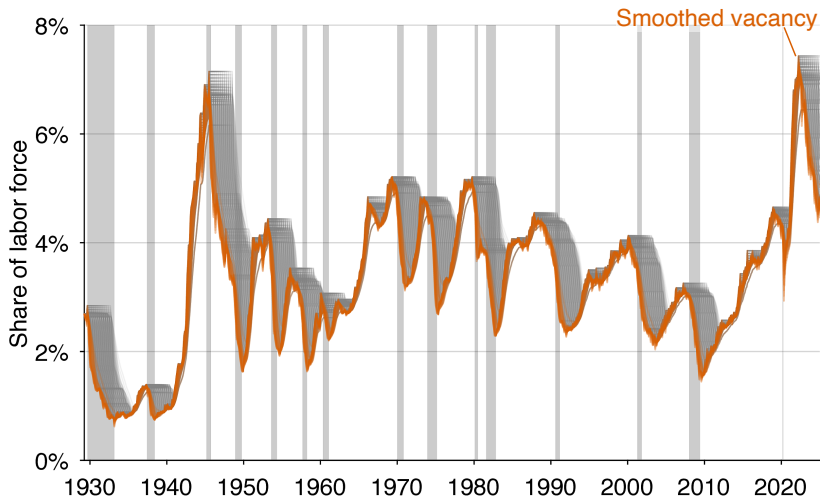
DETECTING UNEMPLOYMENT TURNING POINTS



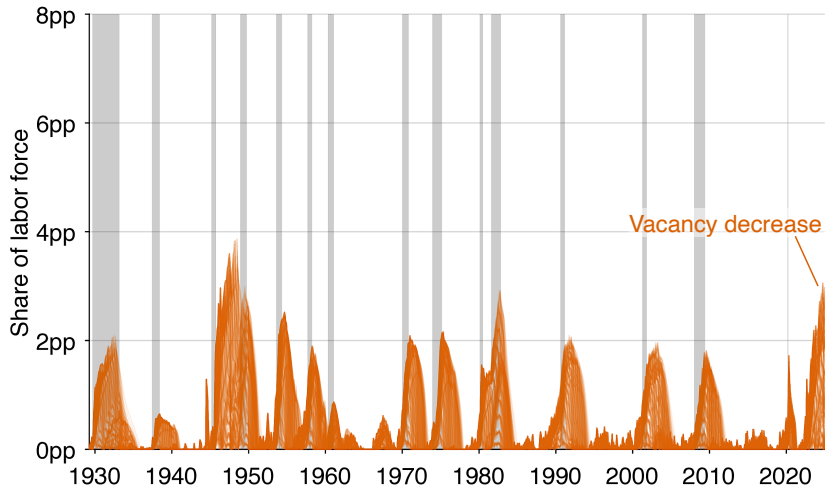
DETECTING VACANCY TURNING POINTS



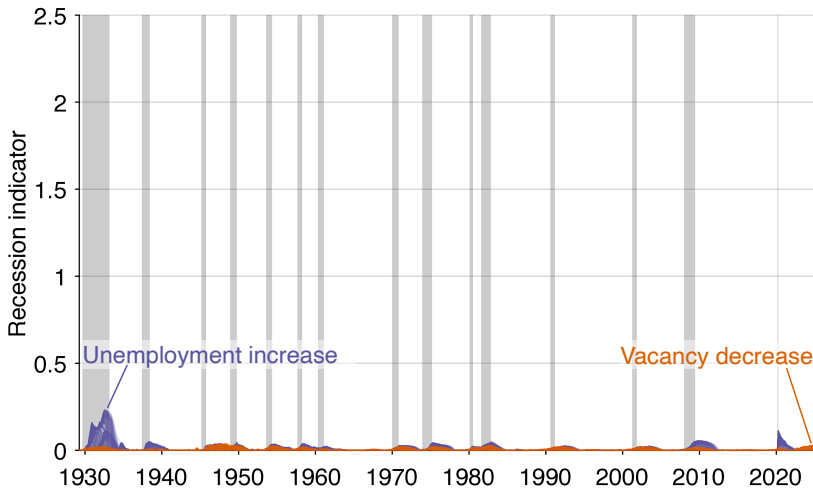
DETECTING VACANCY TURNING POINTS



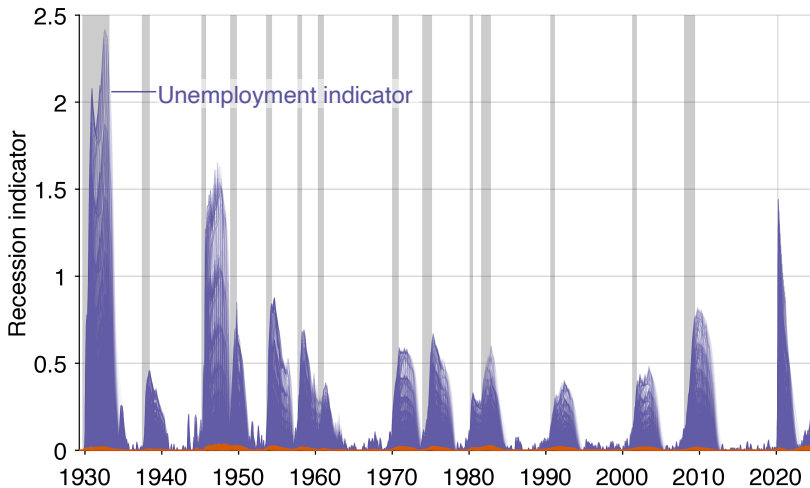
DETECTING VACANCY TURNING POINTS



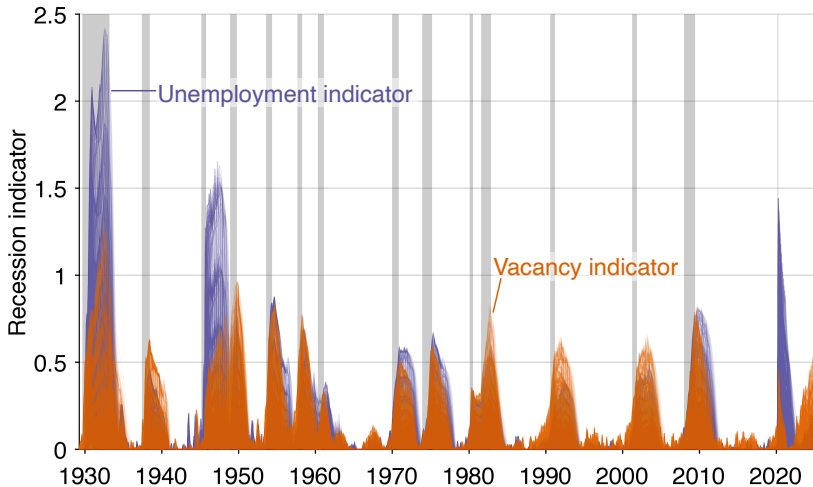
CURVING UNEMPLOYMENT & VACANCY INCREASES



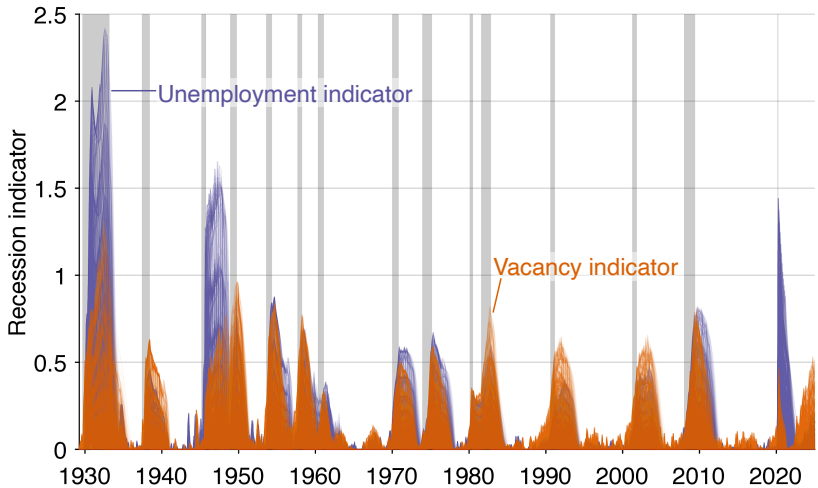
CURVING UNEMPLOYMENT & VACANCY INCREASES



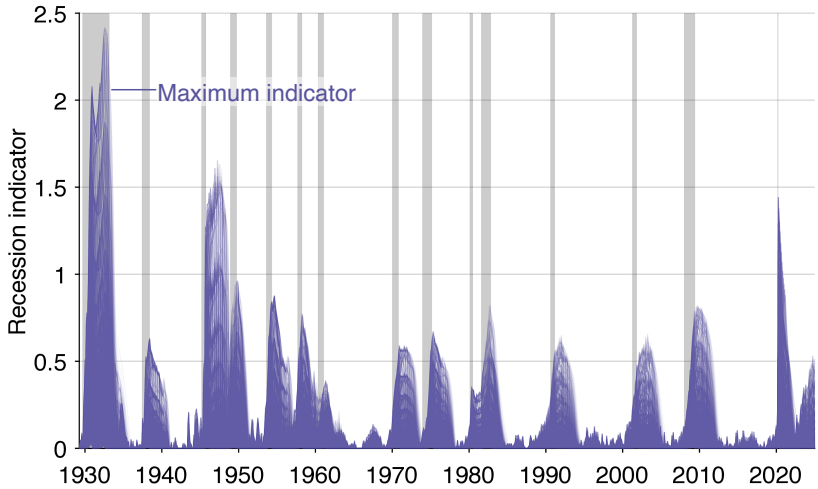
CURVING UNEMPLOYMENT & VACANCY INCREASES



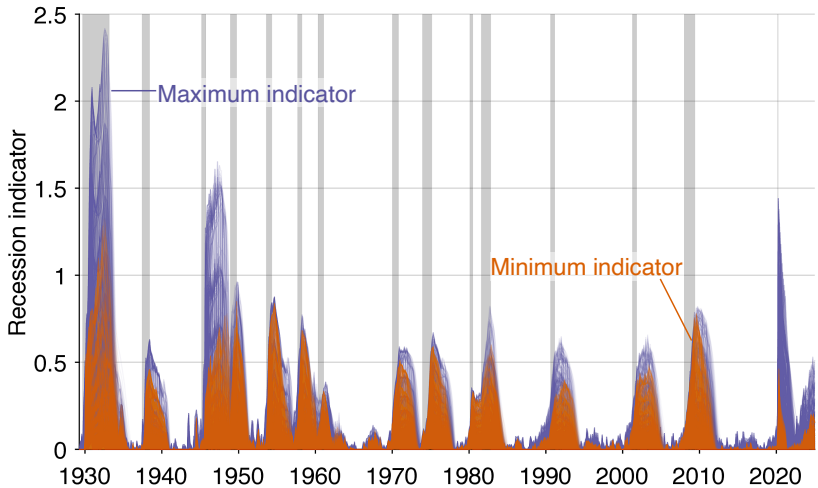
COMBINING UNEMPLOYMENT & VACANCY INDICATORS



COMBINING UNEMPLOYMENT & VACANCY INDICATORS

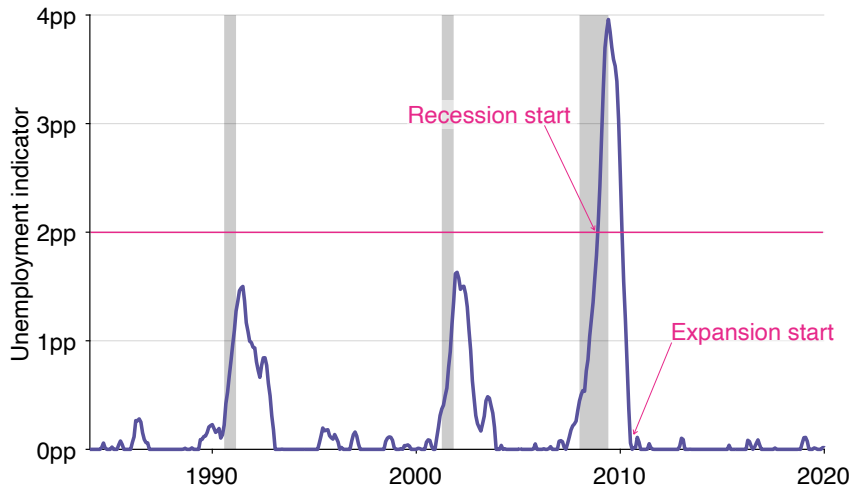


COMBINING UNEMPLOYMENT & VACANCY INDICATORS

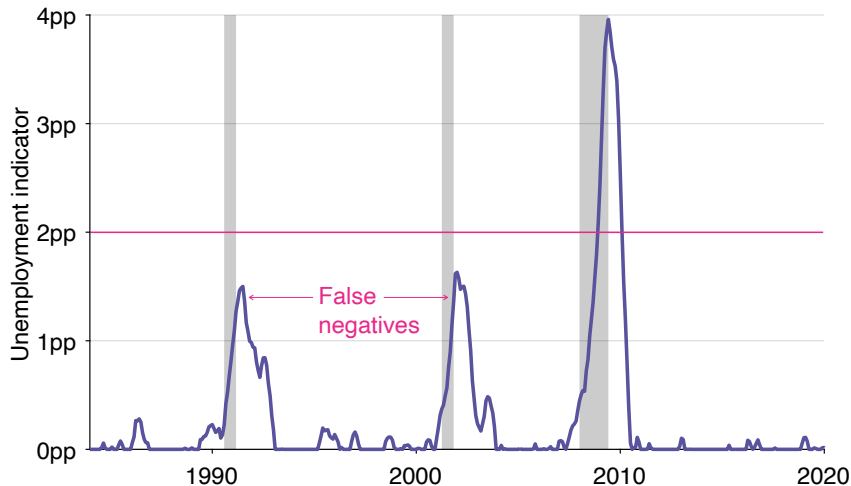


TRAINING RECESSION CLASSIFIERS

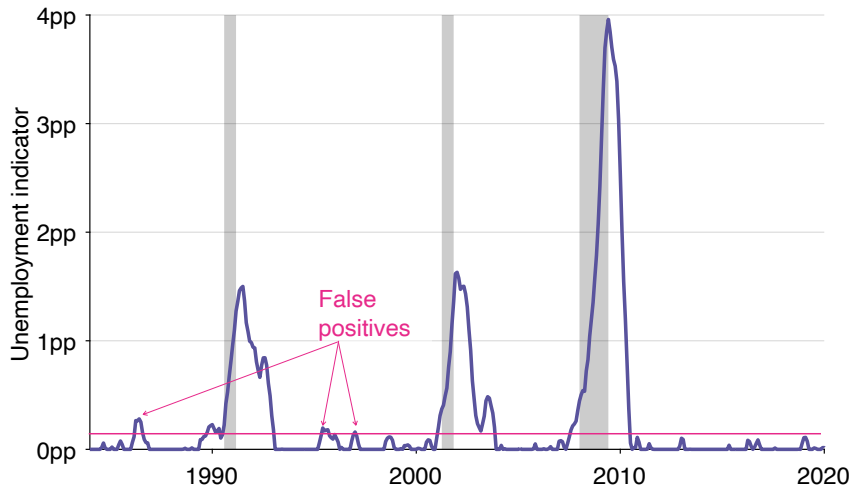
RECESSION CLASSIFIER = INDICATOR + THRESHOLD



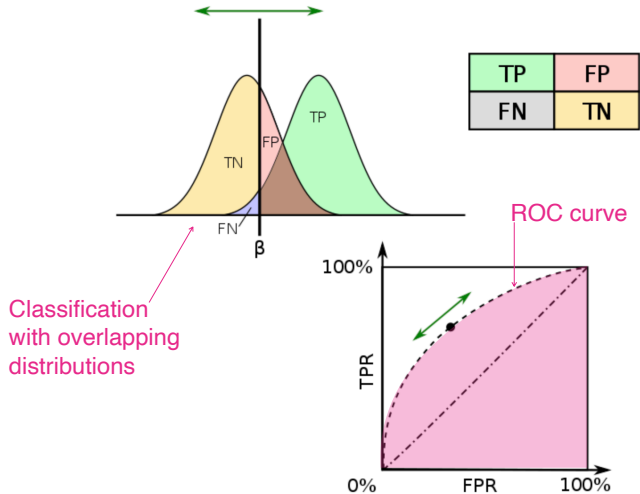
TRAINED CLASSIFIERS HAVE NO FALSE NEGATIVES...



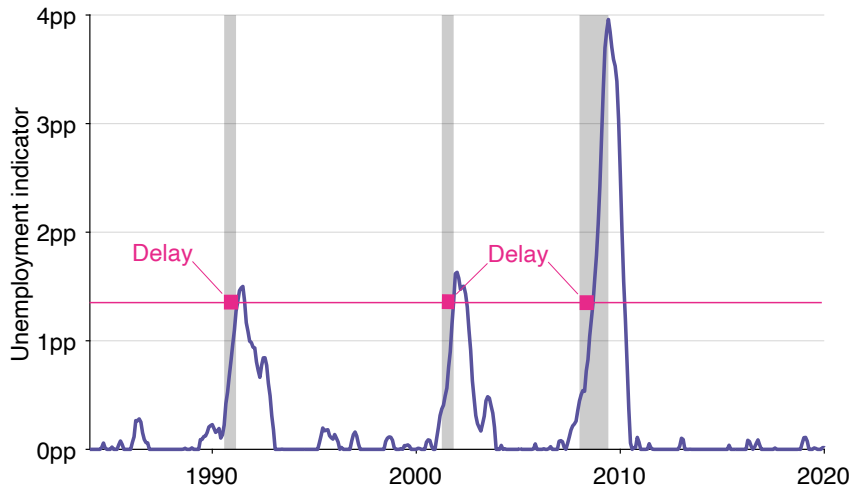
...AND NO FALSE POSITIVES



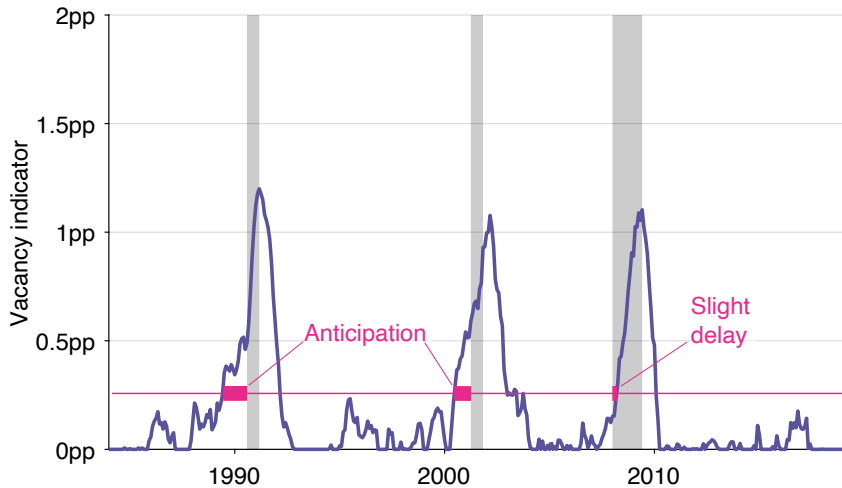
THE TYPICAL METHOD TO RANK CLASSIFIERS DOESN'T APPLY HERE: TOO MANY PERFECT CLASSIFIERS



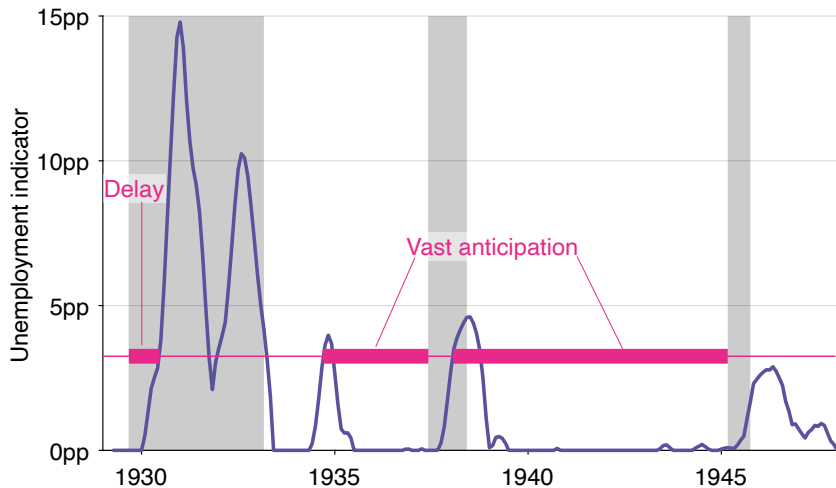
TRAINED CLASSIFIER WITH DELAY



TRAINED CLASSIFIER WITH HIGHER ANTICIPATION

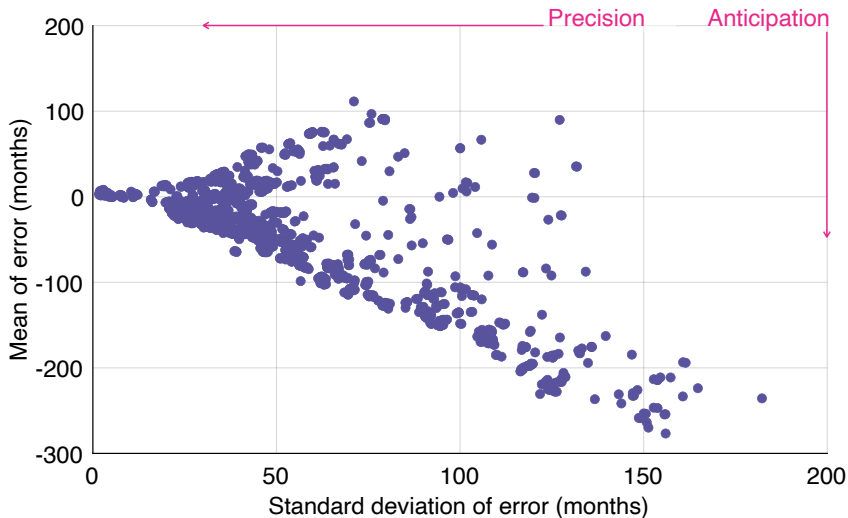


TRAINED CLASSIFIER WITH LOW PRECISION

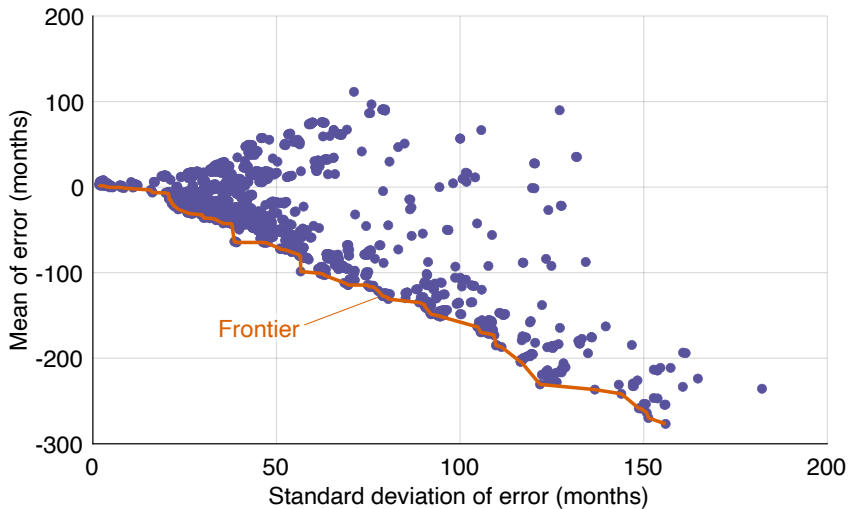


SELECTING RECESSION CLASSIFIERS

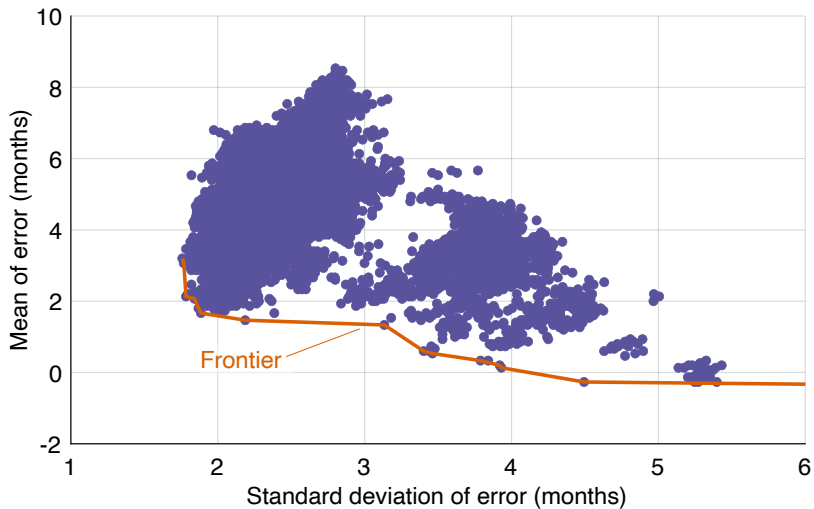
ANTICIPATION-PRECISION OF 2,343,752 CLASSIFIERS



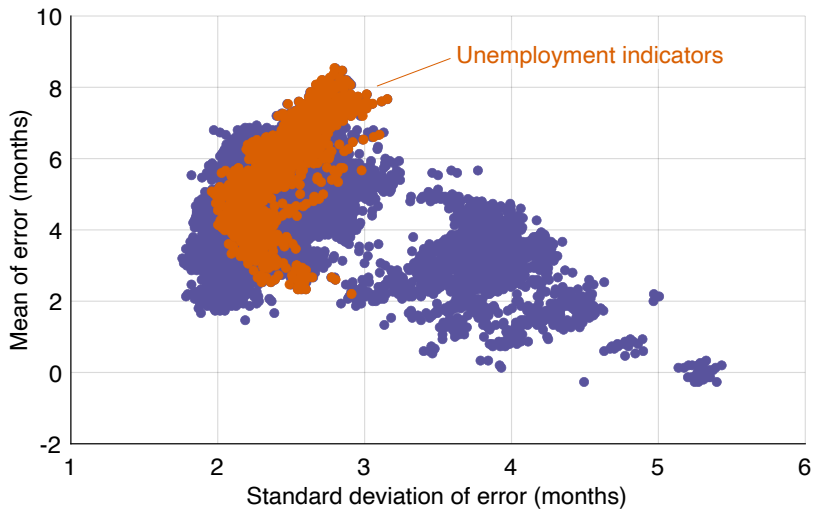
ANTICIPATION-PRECISION FRONTIER: 210 CLASSIFIERS



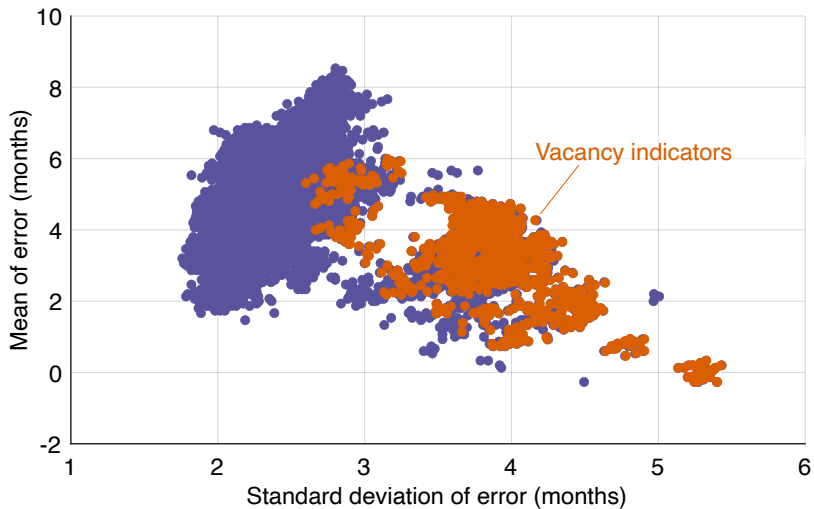
GENERALIZATION OF MICHEZ-RULE RESULTS



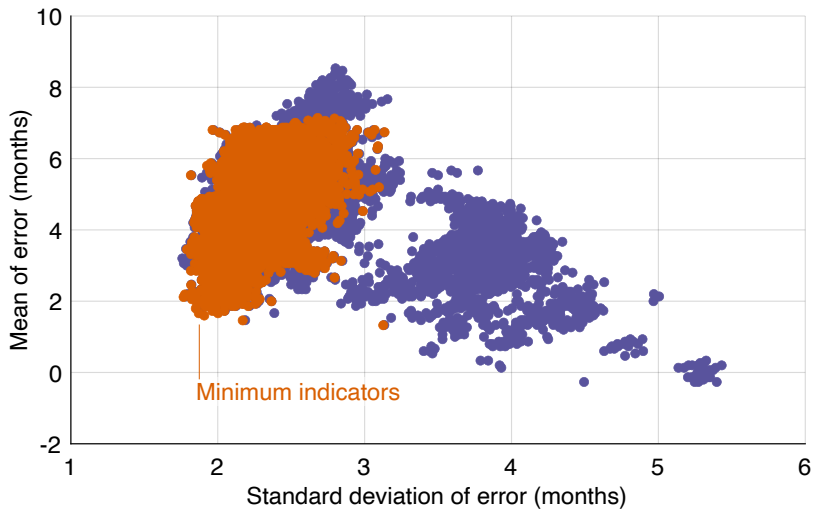
GENERALIZATION OF MICHEZ-RULE RESULTS



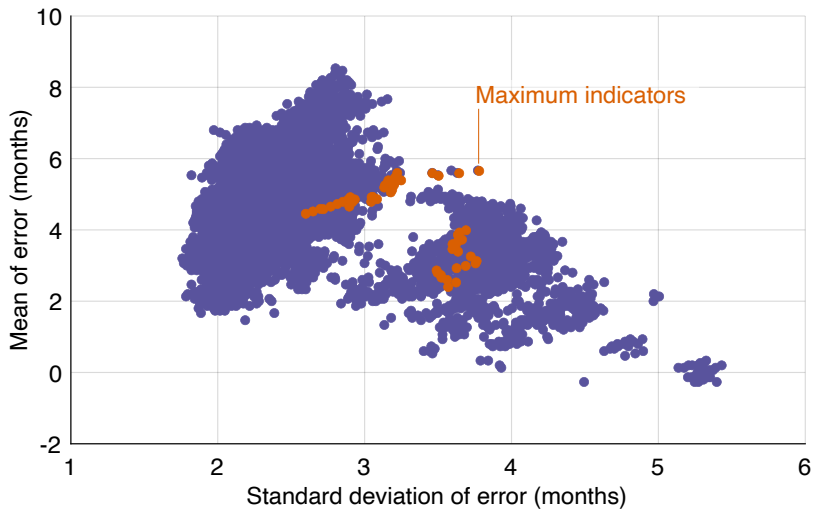
GENERALIZATION OF MICHEZ-RULE RESULTS



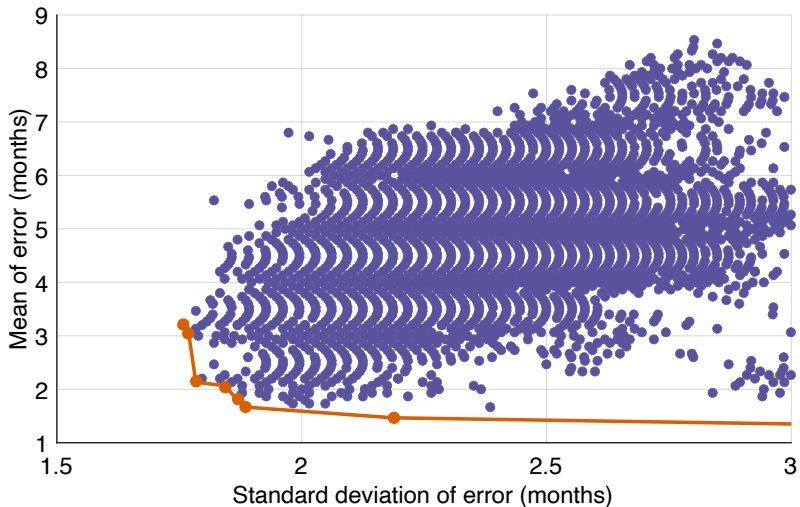
GENERALIZATION OF MICHEZ-RULE RESULTS



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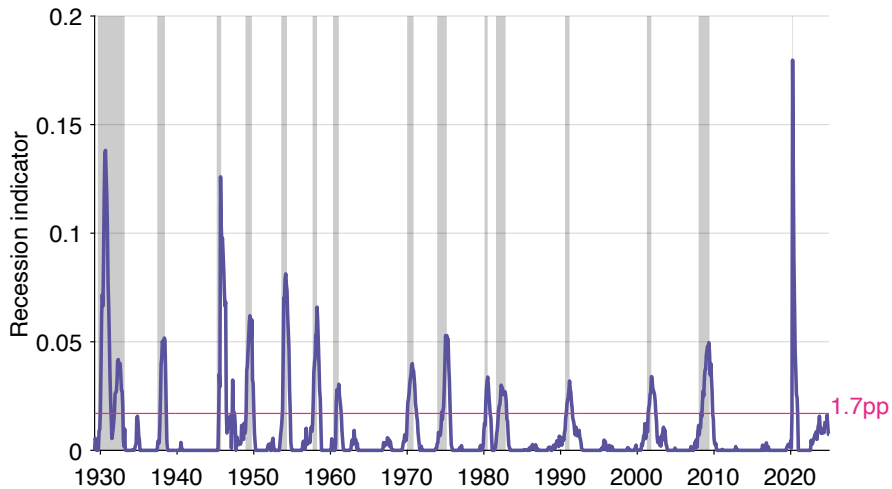
DETECT RECESSION WITHIN 1 YEAR: 7 FRONTIER CLASSIFIERS



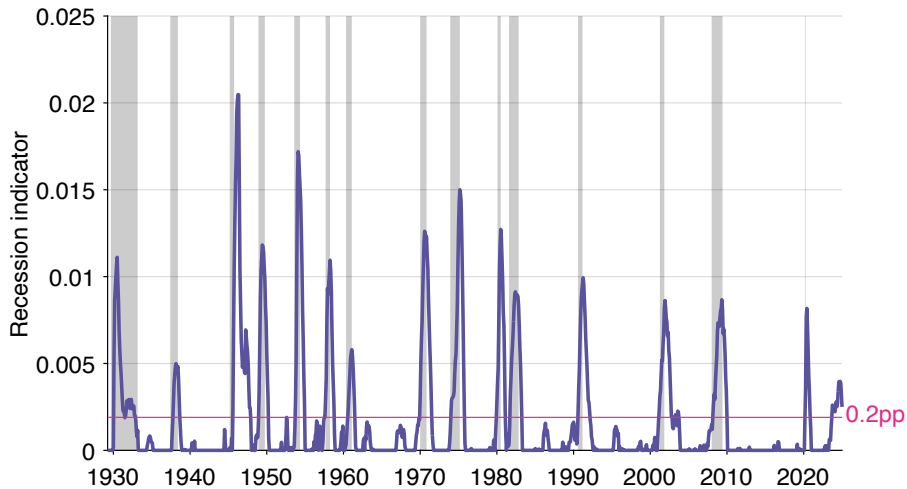
SOME CLASSIFIERS ON THE FRONTIER

	Classifier A	Classifier B	Sahm
Smoothing method	simple	exponential	simple
Smoothing horizon α	8m	0.4	3m
Turning horizon β	1m	9m	12m
Curving weight γ	log	linear	linear
Combination method	u-v	min-max	u-v
Combination weight δ	0.6	1	1
Threshold ζ	1.7pp	0.2pp	0.5pp
Std deviation of error	1.8m	2.2m	
Mean of error	3.1m	1.5m	

CLASSIFIER A

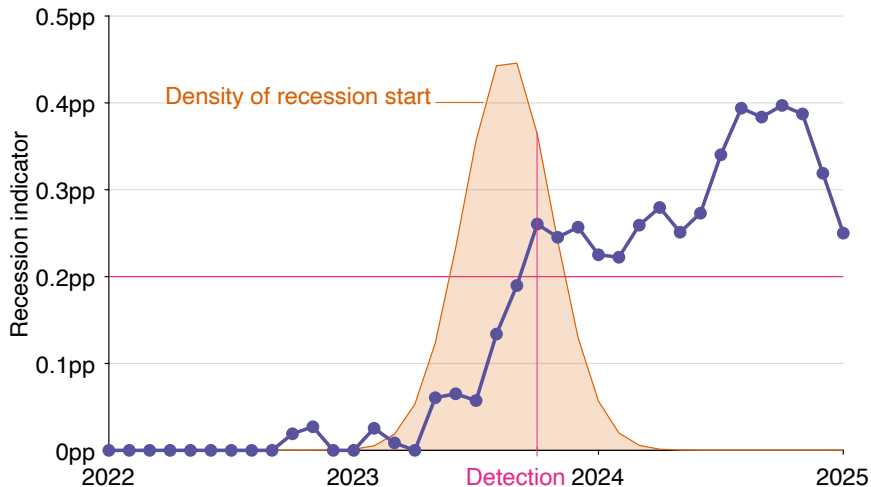


CLASSIFIER B

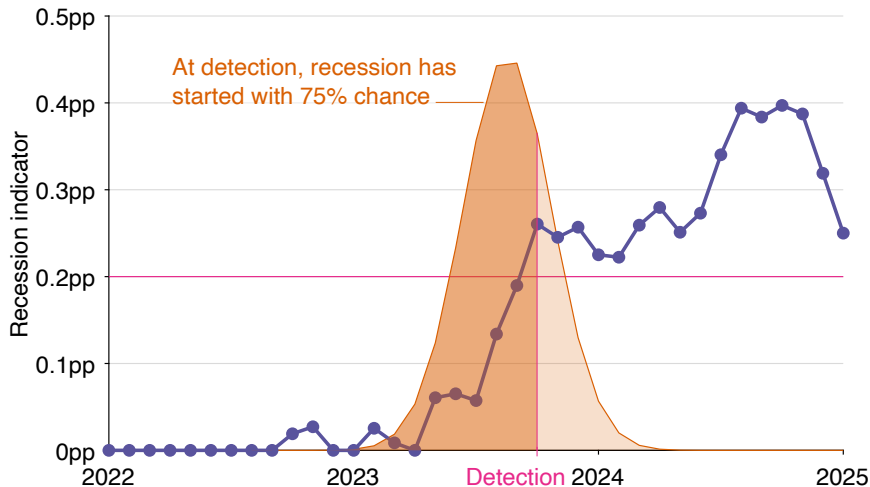


DETECTING US RECESSIONS

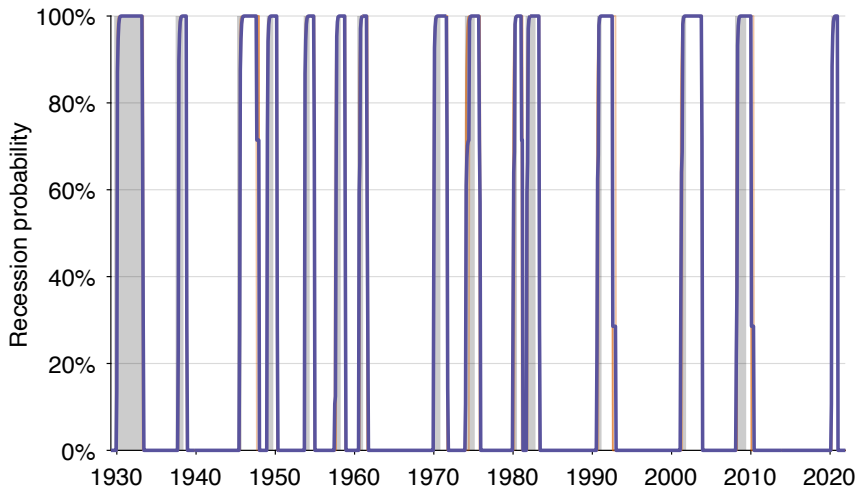
CLASSIFIERS ALSO GIVE RECESSION PROBABILITIES



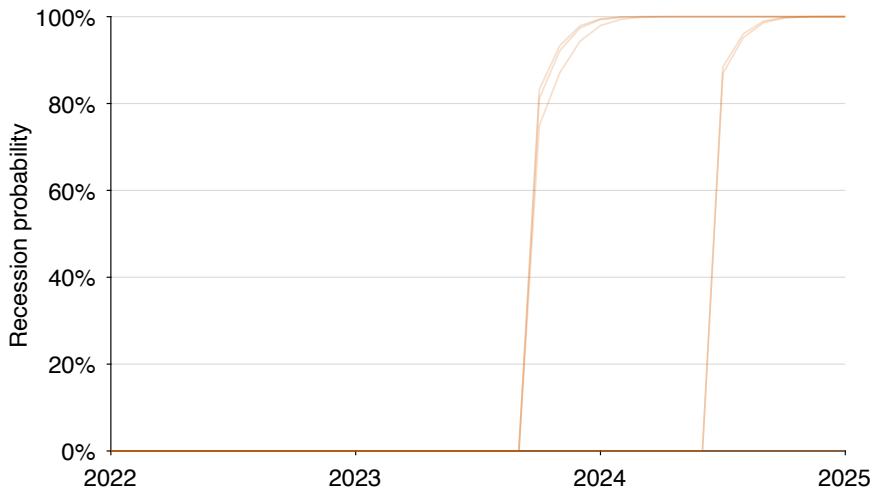
CLASSIFIERS ALSO GIVE RECESSION PROBABILITIES



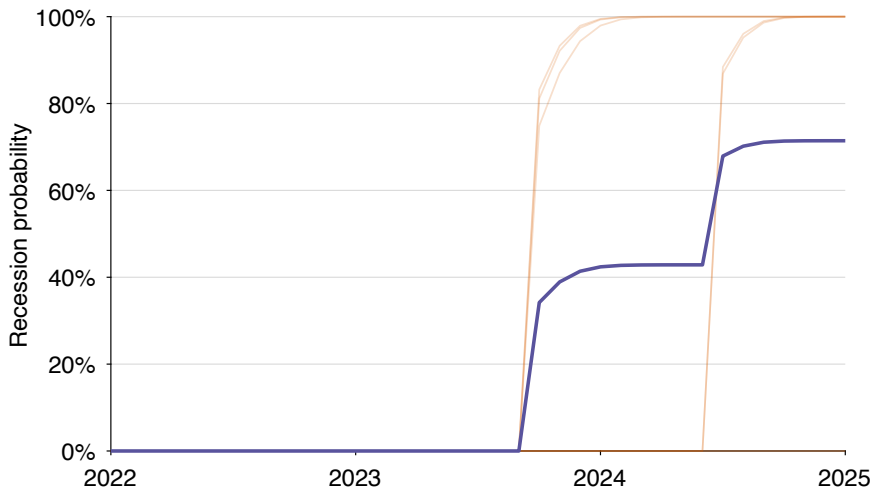
TRAINING RESULTS FOR 7 DETECTION CLASSIFIERS



HAS THE RECESSION STARTED? YES, WITH 71% PROBABILITY

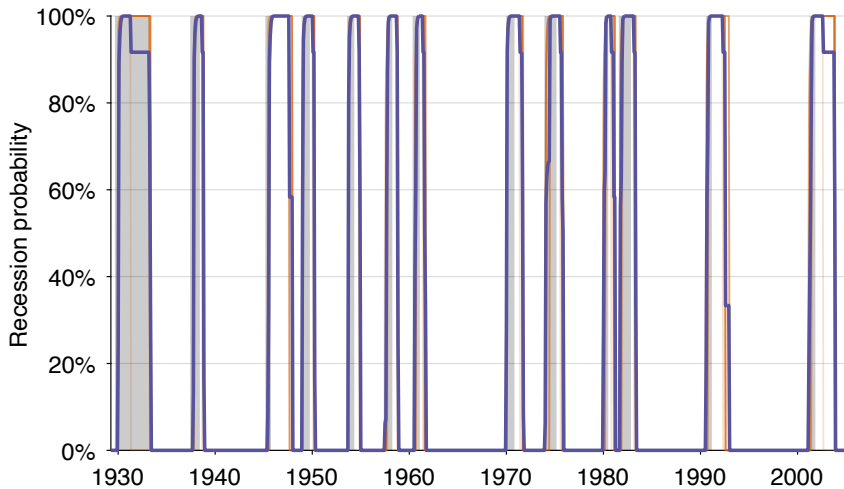


HAS THE RECESSION STARTED? YES, WITH 71% PROBABILITY

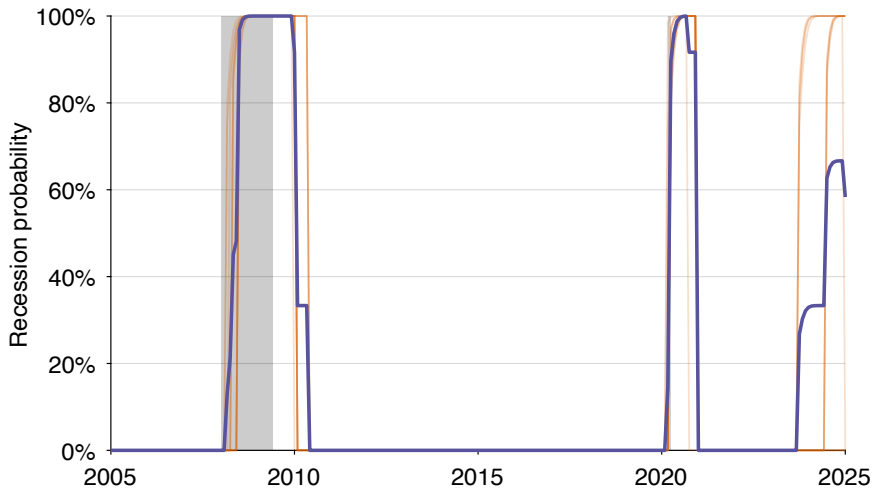


BACKTESTING THE DETECTION METHOD

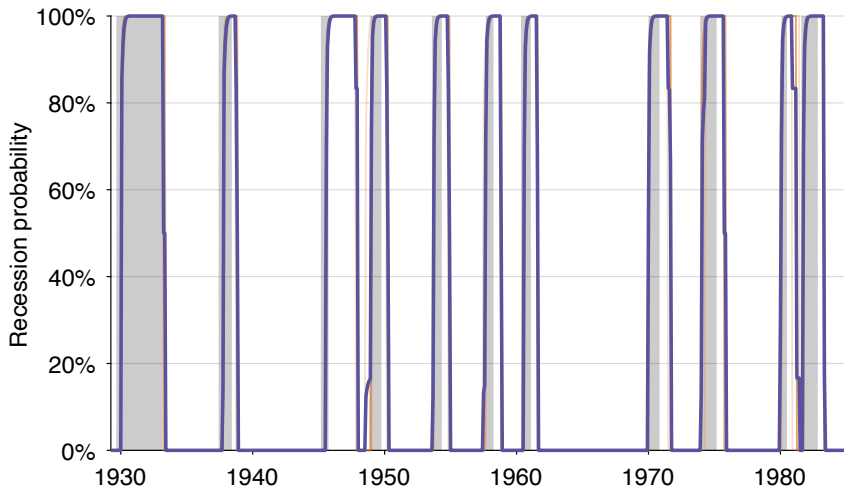
BACKTESTING TO 2005: TRAINING ON 13 RECESSIONS



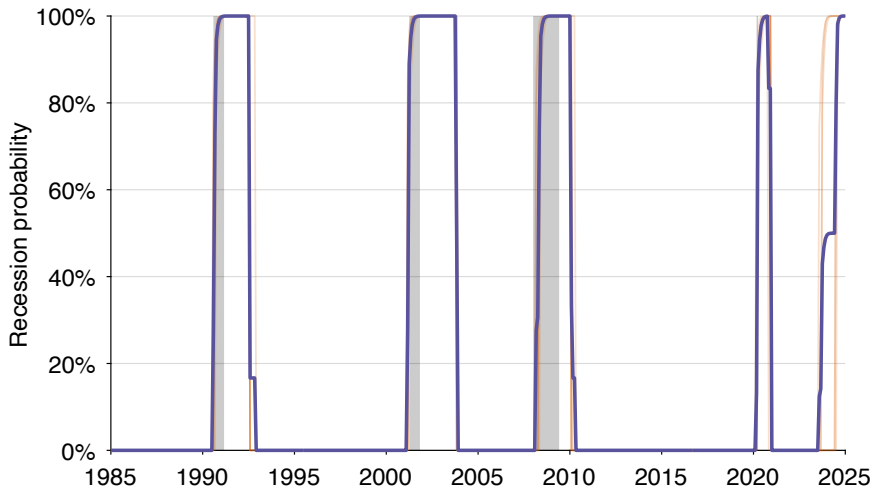
BACKTESTING TO 2005: DETECTING 2+ RECESSIONS



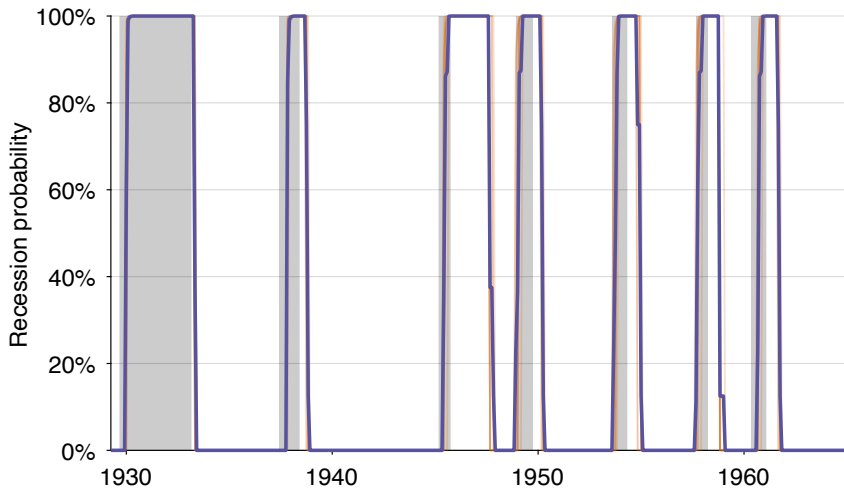
BACKTESTING TO 1985: TRAINING ON 11 RECESSIONS



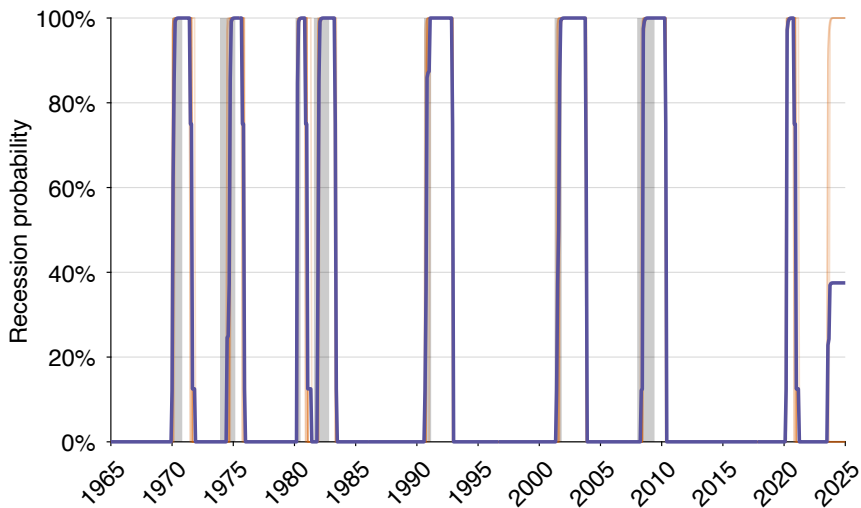
BACKTESTING TO 1985: DETECTING 4+ RECESSIONS



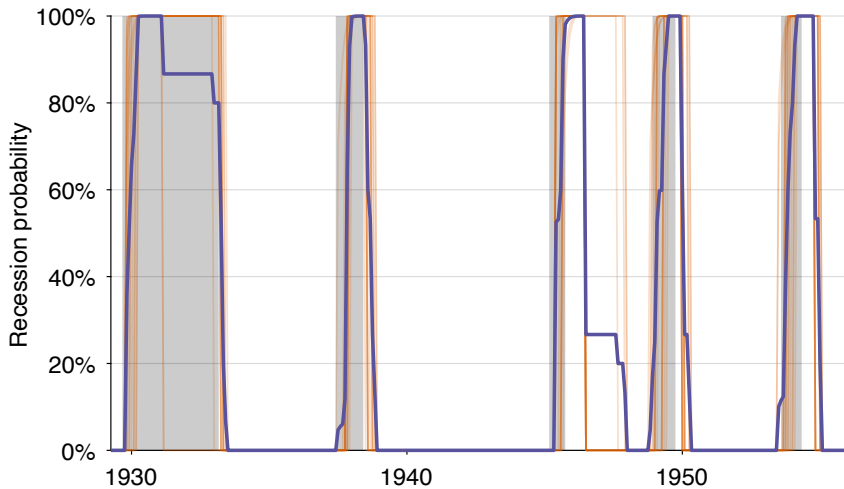
BACKTESTING TO 1965: TRAINING ON 7 RECESSIONS



BACKTESTING TO 1965: DETECTING 8+ RECESSIONS



BACKTESTING TO 1956: TRAINING ON 5 RECESSIONS



BACKTESTING TO 1956: DETECTING 10+ RECESSIONS

